# NLP-DL One Pager-

## Toxic Comment Moderation for Low-Resource Languages

### Team Information

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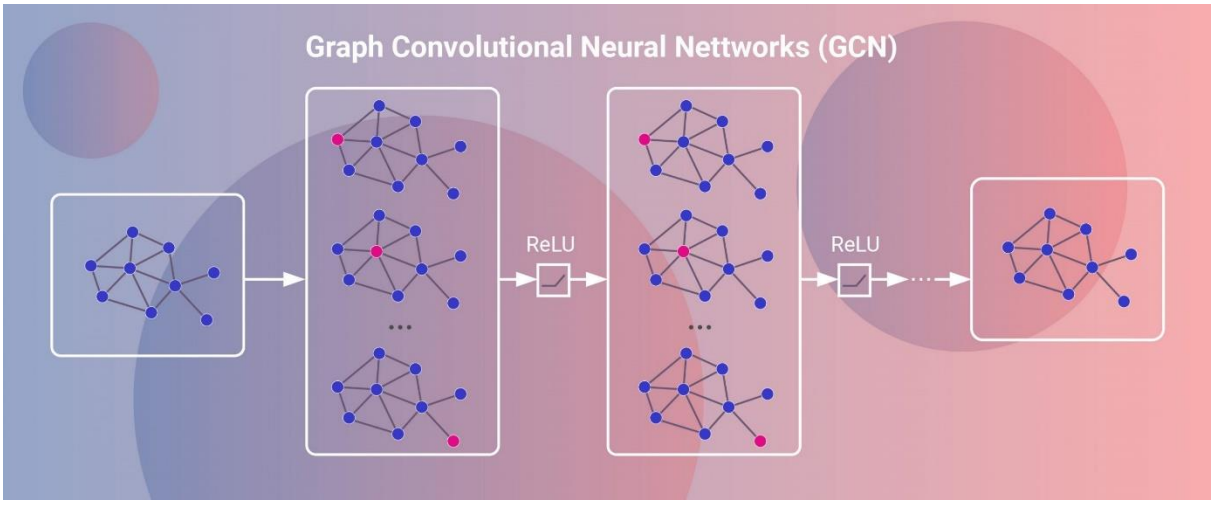
### Abstract

This project develops a robust toxic comment moderation framework tailored for low-resource languages. Unlike traditional toxicity detection systems that rely heavily on large annotated corpora and standardized linguistic structures, the proposed system is designed to operate effectively under limited data availability, lexical variability, and informal user-generated content.

The framework focuses on identifying toxic, abusive, hateful, or offensive content in low-resource language settings where annotated datasets, pretrained models, and linguistic tools are scarce. The system integrates domain-adaptive preprocessing, data augmentation strategies, and fine-tuned transformer-based models to improve generalization under data scarcity.

A multilingual pretrained language model is fine-tuned for toxicity classification using carefully curated low-resource datasets. To address sparse supervision, the pipeline incorporates techniques such as transfer learning, cross-lingual embedding alignment, weak supervision, and class imbalance handling. The system follows a preprocessing → representation learning → fine-tuning → evaluation pipeline.

The central hypothesis is that combining transfer learning with targeted noise handling and low-resource adaptation strategies significantly improves toxic comment detection performance compared to naïve fine-tuning approaches, while maintaining interpretability and robustness in noisy social media environments.




# Uniqueness

## Low-Resource-Centric Toxicity Modeling

Most toxicity detection systems are developed for high-resource languages such as English. This project explicitly focuses on low-resource languages where labeled data is scarce, spelling variations are common, and linguistic tools are limited. The modeling decisions are tailored specifically for such constraints rather than adapting high-resource assumptions.

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## Transfer Learning with Domain Adaptation

Instead of training models from scratch, this framework leverages multilingual pretrained transformers (e.g., mBERT, XLM-R) and adapts them to the target low-resource language domain. Domain-specific fine-tuning and augmentation techniques are incorporated to reduce overfitting and improve generalization.

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## Robustness to Noisy Social Media Text

Low-resource online text often contains:

- spelling variations
- slang and informal grammar
- emojis and hashtags
- code-mixing
- character elongations

The proposed system includes preprocessing and normalization strategies specifically designed for handling such noise.

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## Class Imbalance and Sparse Supervision Handling

Toxic comment datasets are often highly imbalanced (more non-toxic than toxic examples). The system incorporates:

- weighted loss functions
- focal loss
- data augmentation
- synthetic minority oversampling

to ensure fair learning across classes.

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## Interpretable Moderation Decisions

The framework explores attention visualization and token-level importance scoring to understand which words contribute to toxicity predictions. This helps ensure transparency and supports explainable AI principles in content moderation.

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## Low-Resource Language Setting

Low-resource languages are characterized by limited digital corpora, few annotated datasets, lack of standardized benchmarks, and minimal NLP tooling. Unlike English or other high-resource languages, these languages often lack large-scale toxic comment datasets and pretrained domain-specific models.

Social media content in such languages is often:

- Informal
- Orthographically inconsistent
- Code-mixed with dominant languages
- Rich in slang and region-specific expressions

The scarcity of labeled data creates challenges such as:

- Overfitting during training
- Poor generalization
- Difficulty detecting implicit or context-dependent toxicity

This project addresses these challenges by leveraging multilingual pretrained models, transfer learning, and data-efficient fine-tuning strategies. Instead of relying purely on large datasets, the approach focuses on maximizing performance under constrained supervision.

By studying toxicity detection in low-resource settings, the project contributes toward building inclusive AI systems that extend content moderation support beyond dominant global languages.

# Learning Goals

## 1. Low-Resource Representation Learning

Develop a toxic comment classification model using multilingual pretrained embeddings and fine-tuning strategies suitable for limited data environments.

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## 2. Handling Noisy User-Generated Content

Design preprocessing techniques to handle:

- spelling normalization
- emojis and hashtags
- character repetition
- code-mixing
- slang and informal variations

to improve model robustness.

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## 3. Model Comparison and Baseline Analysis

Implement and compare:

- Traditional machine learning baselines (Logistic Regression, SVM)
- Deep learning models (LSTM, BiLSTM)
- Transformer-based models (mBERT, XLM-R)

Evaluate using:

- Accuracy
- Precision

- Recall
- F1-score
- Macro-F1 (for imbalance handling)

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## 4. Low-Resource Training Strategy

Investigate performance under:

- Reduced training data
- Imbalanced class distributions
- Cross-domain testing
- Few-shot learning scenarios

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## 5. Explainability in Toxic Comment Detection

Analyze:

- Attention weights
- Token importance
- Error patterns

to understand model behavior and reduce unintended bias.

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## Lit Survey

| Sl. No | Title | Link to Paper |
|---|---|---|
| 1 | Detoxify: A Large-Scale Multilingual Toxicity Detection | ylmmcl at Multilingual Text Detoxification 2025: Lexicon-Guided Detoxification and Classifier-Gated Rewriting |

| | | |
|---|---|---|
| 2 | Leveraging Deep Learning for Comprehensive Multilingual Hate Speech Detection | Hate speech detection in low-resourced Indian languages: An analysis of transformer-based monolingual and multilingual models with cross-lingual experiments \| Natural Language Processing \| Cambridge Core |
| 3 | Multilingual BERT for Cross-Lingual Transfer | Syntax-augmented Multilingual BERT for Cross-lingual Transfer |
| 4 | XLM-R: Unsupervised Cross-Lingual Representation Learning | Unsupervised Cross-lingual Representation Learning at Scale |
| 5 | Data Scarcity in NLP | Overcoming Data Scarcity in Generative Language Modelling for Low-Resource Languages: A Systematic Review |
| 6 | Focal Loss for Dense Object Detection | Focal Loss for Dense Object Detection — dl 0.0.1 documentation |
| 7 | Transfer Learning in Natural Language Processing: Overcoming Low-Resource Challenges | thejaswi-adimulam-swetha-chinta-suprit-kumar-pattanayak_ImVmG.pdf |

## Datasets (with Links)

**Multilingual Toxic Comment Classification Dataset:**

Link: https://github.com/AkshayaANANTAM/Multilingual-Toxic-comments-classification

**Hate Speech and Offensive Language Datasets (Low-Resource Variants)**

Dataset using Twitter data, is was used to research hate-speech detection. The text is classified as: hate-speech, offensive language, and neither. Due to the nature of the study, it's important to note that this dataset contains text that can be considered racist, sexist, homophobic, or generally offensive.

Link: Hate Speech and Offensive Language Dataset

**Cyberbullying Dataset**

This dataset is a collection of datasets from different sources related to the automatic detection of cyber-bullying. The data is from different social media platforms like Kaggle,

Twitter, Wikipedia Talk pages and YouTube. The data contain text and labeled as bullying or not. The data contains different types of cyber-bullying like hate speech, aggression, insults and toxicity.
Link: Cyberbullying Dataset